

DataTürk

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Abstract—This document is the first progress report for DataTürk’s machine learning project. The document contains the details of the dataset cleansing process and the summary of the given dataset.

Index Terms—machine learning, github, dataset, missing values, graphs

I. INTRODUCTION

We have created a python script¹, in order to identify the missing values in our given dataset, and analyze the data according to their contextual meaning. Later, we visualised some parts of the dataset to be able to understand it better.

II. DATA PREPROCESSING

A. Dropping Some of the Columns and Filling the Values

We identified the columns that contained more than 50 percent of the missing data values in them and filled in these numeric columns using the calculated mean of such columns [1]. For non-numeric columns, we used the mode. In order to eliminate the bugs in the code, we took advantage of some sources [2] [3] [4].

B. Creating a Binary Label for the Dataset

We converted the last column to a binary label column, which gave us a way to tag and identify these columns.

III. APPENDICES

A. Summary of the Dataset

The dataset is can be seen as:

Statistic	Value
Missing values (before)	7318445
Non-numerical columns	13
Numerical columns	68
Total Rows	246008
Total Columns	81

¹<https://github.com/iamsahinemir/DataTürk>

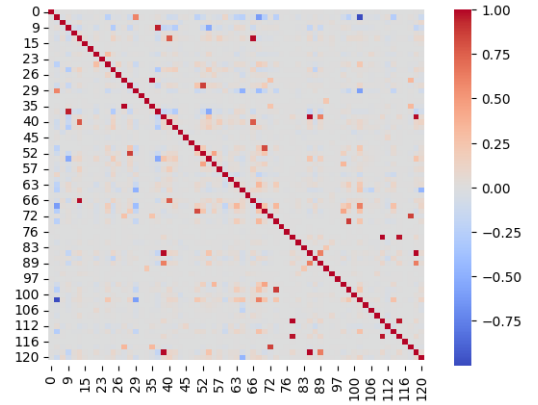


Fig. 1. Correlation between information on the dataset.

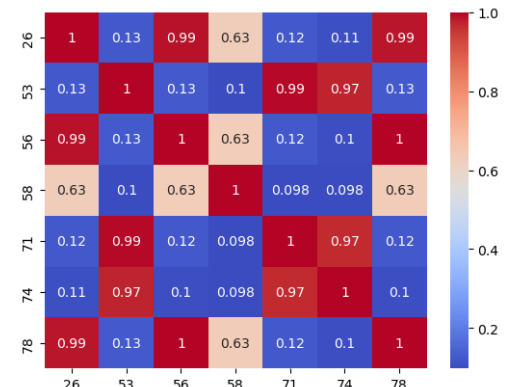


Fig. 2. Heatmap.

Correlation analysis was performed in the Fig. 1. Our main purpose while doing this analysis is to determine the relationship between the variables and to understand them according to their directions and intensities. In this statistical

method, it can be said that there is a significant relationship between both variables if it is less than 0.05 among the meanings we can understand. In our analysis, we proved this again in the heat map above, where the number values are the same in the red visible line. The reason for using heatmap is actually to see the losses, separations or densities more clearly again.

- [2] Güneş, A. Python ile Veri Temizleme ve Ön İşleme. [Online]. Available: <https://alprgunes.medium.com/>
- [3] TirendazAkademi. Nasıl Veri Bilimcisi Olunur. [Online]. Available: <https://tirendazakademi.medium.com/>
- [4] OpenAI. ChatGPT-3. [Online]. Available: <https://chat.openai.com/>

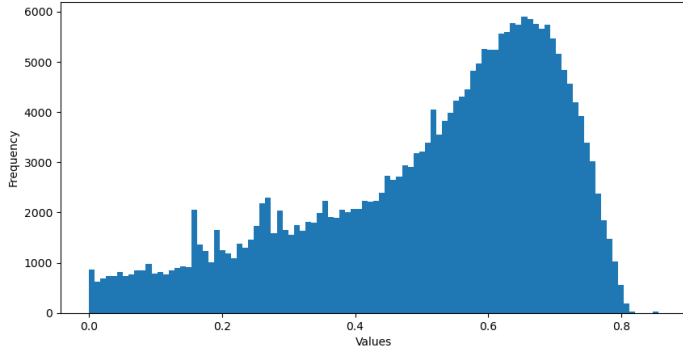


Fig. 3. Histogram of Column 15.

What can be seen is the aim aimed in the histogram Fig. 3 is actually used to determine the frequency distribution. As can be seen, it is observed that the 15th column is tilted to the right. As can be understood, we see that the value of 0.6 in milk 15 is more common. Among the reasons that we can see as the reasons for not being symmetrical will be the inconsistency of the variables.

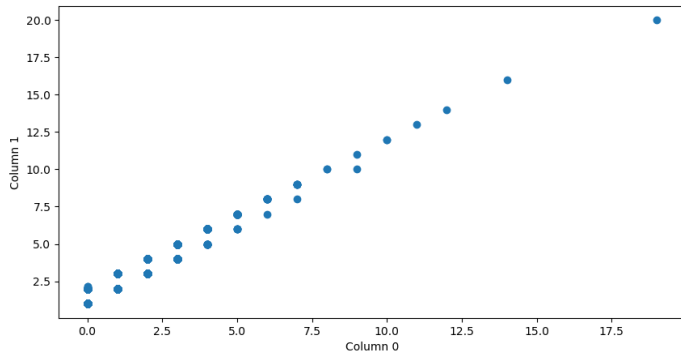


Fig. 4. Positive correlation between the column 50 and 64.

The purpose of the Scatter Plot, which is used for columns 50 and 64, is to examine the relationship between two variables and to detect the change between the variables corresponding to two variables. As a result, a positive and strong relationship emerged.

IV. CONCLUSION AND FUTURE WORK

Our next step will be to go on the depths of machine learning with our cleansed dataset.

REFERENCES

- [1] Cook, A. Handling Missing Values. [Online]. Available: <https://www.kaggle.com/code/alexisbcook/handling-missing-values>